# Small Data Domain Generalization for Cross-Platform Sentiment Classification

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## 1 Introduction

Machine learning models are typically trained and evaluated under the assumption that training and test data are drawn from the same data distribution. However, in real-world applications, this assumption often fails. Data collected in different contexts, time periods, or formats can exhibit significant variations, a phenomenon known as domain shift. As a result, models that perform well on training data may perform poorly when deployed in unseen settings. Domain generalization addresses this challenge by aiming to learn models that generalize well even on unseen domains, without access to target-domain data during training. To achieve this, domain generalization focuses on learning invariant features, which are robust and transferable representations from training domains that capture the underlying structure of the task rather than spurious correlations, which are superficial domain-specific features that do not hold across environments. In this project, we study domain generalization in data-scarce settings across multiple online platforms. We focus on building a sentiment classifier across different online platforms as a base task, evaluating whether models trained on limited data from certain domains can generalize to unseen platforms with different language styles. As an extension, we explore the more challenging task of detecting AI-generated reviews under domain shift.

## 2 Workshop Challenge Link

This project is inspired by the CVPR 2026 Workshop: *Domain Generalization: Evolution, Breakthroughs, and Future Horizons*. The link to the challenge is: workshop website.

## 3 Approach

For this project, we study domain generalization in cross-platform text classification using a leave-one-domain-out evaluation setup across four domains: Twitter (X), Amazon, Yelp, and Reddit. These domains differ significantly in formality, writing style, text length, and topics. For example, Yelp and Amazon reviews are generally more structured and formal, while Twitter and Reddit contains shorter text and more informal language, including slang, abbreviations, and emojis. These differences make them suitable for studying how well models generalize under domain shift.

To study domain generalization under limited data, we conduct experiments using 100%, 50%, 25%, and 10% of the labeled data. This allows us to evaluate how the amount of data and domain shift affects performance. We use a pretrained transformer-based text encoder, such as BERT or DistilBERT, as our backbone. We will also implement several domain generalization techniques to improve cross-domain robustness. First, we will use balanced sampling, which ensures that training batches contain a more even distribution of examples across source domain to reduce selection bias of the model. Second, we apply text augmentation methods to increase linguistic diversity in the data. This includes replacing with synonyms or paraphrasing. Third, text normalization such as lower-casing and handling emoji characters and spellings will be used to reduce variation across domains.

Preprint.

Beyond these pre-processing strategies, we will also explore domain adversarial training (DANN) (Ganin et al., 2016), a domain generalization-specific method. In this setup, the model consists of a shared text encoder, a sentiment classification head, and a separate domain classification head. The sentiment head learns to predict sentiment labels, while the domain head attempts to predict the source platform. Through adversarial training, the encoder is encouraged to learn representations that are useful for sentiment classification but less informative about domain identity. In other words, the model is pushed to learn domain-invariant features that transfer better to unseen platforms. Lastly, we will also explore probabilistic feature mapping since it had a high performance and robustness for small data in "Domain Generalization with Small Data"(Chen et al., 2024)

## 4 Motivation

We were motivated to do this project because domain shift is a real problem when it comes to text classification in the real world. A model might perform very well on one platform but fail when used on another platform, since the writing style, slang, structure, or even tone could be completely different depending on what website you are looking at. Since online text comes from everywhere across the internet, it is very important that a model can work well no matter which platform it is getting its data from.

We are also very interested in this project because of the possibility of applying these same ideas to AI-generated text detection. With AI-generated text becoming more and more common across the internet, it has made its way into our everyday lives, from reviews and comments to even entire posts. Because of that, being able to tell whether a text is written by a human or generated by AI has become increasingly important now more than ever. The problem is that it is extremely difficult to tell if a text is AI-generated, as AI text can appear across multiple different platforms and styles, which is the same domain shift problem we are tackling.

This project gives us the ability to study both of these issues at the same time. First, it helps us understand how to build text classifiers that remain stable even when the text platform changes. Second, it gives us a way to explore a practical application of domain generalization in AI text detection. We think this problem matters a lot because stronger cross-domain models could help NLP systems perform more consistently when used in the real world.

## 5 Data and Code

We will use publicly available datasets from multiple online platforms to study cross-platform sentiment classification under domain shift. For Yelp, we will use the Yelp dataset from Kaggle: Yelp Dataset. We will specifically use the review dataset labeled yelp_academic_dataset_review.json, which comes from Yelp's open academic dataset.

For both Twitter and Reddit, we will use the Twitter and Reddit Sentiment Analysis Dataset. This is useful for our project because it gives us data from two different social media platforms from the same source. The dataset includes about 163K tweets and 37K Reddit comments.

For Amazon, we will use the Amazon Polarity Dataset from Hugging Face. The dataset has 4 million total rows and is already split into 3.6M for training and 400K for test examples. This gives us a large review domain with a very different writing style from Yelp, Reddit, and Twitter.

Since this project is inspired by the workshop, specifically the CVPR 2026 Workshop: *Domain Generalization: Evolution, Breakthroughs, and Future Horizons* (workshop link), we do not have any workshop starter code. We will implement the models ourselves using standard NLP tools like PyTorch. We may also use standard libraries for text cleaning to make sure all the data is in the same standard format. This setup will allow us to compare a standard transformer baseline to several domain generalization methods, including text augmentation, normalization, and domain adversarial training.

## 6 Baselines and Related Work

Our baseline model is a standard transformer (BERT) trained and evaluated on labeled data from a single domain, providing us a benchmark for in-domain performance. Domain generalization will address the issue of training models to perform well on unseen domains by having it learn invariant features that for robust to distribution shifts. Our main approach will include data augmentation and text normalization to reduce differences across domains. Then using DANN to provide a gradient reversal layer to encourage the encoder to learn patterns and representations.

Domain generalization with small data is important for sentiment classification across social media platforms as labeled data is scarce and domain shifts are prevalent. Meta-learning techniques such as MLDG (meta-learning for domain generalization) (Li et al., 2017) trains the model to perform well on out of distribution data by exposing them to domain shifting during the training process, allowing for the learning of general representations of the datasets. When the data is insufficient, we often fall back on probabilistic models that minimizes discrepancy between the source domains. Recent works on domain generalization small datasets propose learning domain-invariant representations through probabilistic frameworks, mapping data points into probabilistic embeddings and using MMD to measure the differences between different distributions. Other approaches uses contrastive semantic alignment to pull positive pairs closer while pushing negative pairs apart, allowing for the learning of semantic structures across domains. Our work takes in these approaches by combining balanced sampling, text augmentation, normalization and adversarial training to evaluate whether models trained on limited data from its source domains are able to generalize to unseen domains.

Our work is purely on domain generalization, as we are limiting the model to a source domain, withholding access to target-domain data during training, then testing whether or not cross-domain sentiment classification can succeed with only 10%-100% of the labeled source data. This differs from domain adaptation, which assumes that the target data is available, and directly aligns with real-world scenarios where the models need to be used on new domains without any retaining or calibration.

# 7 Expected Plan

## 7.1 Achievable Plan

Our goal for this project is to build and train a cross-network sentiment analysis classifier to determine the polarity of text on social media platforms. Our model will train across the four specified domains (Twitter (X), Yelp, Amazon, Reddit) using a varying percentage of the labeled data (100%, 50%, 25%, 10%) in order to study the impact of data scarcity on domain shift and domain generalization.

We will start by using an established baseline: a pretrained BERT/DistiliBERT transformer, which we will then fine tune across the varying percentages of training data. Using a pretrained transformer helps create a benchmark, allowing us to analyze the degree of domain shift between social media domains before the implementation of generalization techniques. Once a benchmark is established, several generalization techniques will be deployed in an attempt to improve our model's cross-domain performance under limited data conditions. We intend to implement balanced sampling to reduce domain selection bias, text augmentation to diversify the training corpus, and text normalization to remove surface-level variation from text or remove noise. These techniques form a realistic and reproducible model training pipeline under our limited-data conditions.

## 7.2 Moon-shot Plan

Our moonshot plan revolves around exploring two additional techniques: domain adversarial training (DANN) and probabilistic feature mapping. These techniques will be applied to our model with respect to time constraints and our individual schedules.

The objective of implementing DANN on our sentiment analysis classifier is to help the model learn domain-invariant representations. The model will be penalized when it relies on domain-specific information, and this is implemented through a domain classification head that the model is training adversarially. DANN will encourage the model to generalize better on both unseen data and unseen domains, which will keep the model robust under limited-data conditions.

The objective of implementing probabilistic feature mapping is to introduce uncertainty in domain-invariant representations. A probability distribution will be modeled across the possible representations, helping our sentiment analysis classifier to be more robust and avoid overfitting on noisy/unreliable features.

## 8 Evaluation

### 8.1 Evaluation Protocol

We opted for a leave-one-domain-out (LODO) protocol to assess domain generalization. For each round, we use one domain as the test domain and train on the remaining source domains. This setup makes sure that the model has never seen the labeled examples from the target domains during training, testing whether domain generalization help generalize better on unseen domains.

For each source domain, we sample the labeled data at 100%, 50%, 25%, 10% to evaluate how the data reduction relates to the domain shift. With each limit, we train the model using the techniques described previously. We report the performance separately for each split and then combine the results across all splits to obtain the mean performance.

### 8.2 Metrics

We will report the accuracy as the primary metric, as it is the standard for sentiment classification problems and provides a clear measurement of how well the model generalizes across platforms. We will additionally incorporate an F1-score (macro average) to account for potential class imbalances across domains. For each data limit, we will report:

- **In-domain Accuracy:** Performance on test sets from the source domains used during training (Measures how well the model performs on seen domains as we reduce training data)
- **Cross-domain accuracy:** Performance on held-out target domain (Measures the generalization to unseen platforms)
- **Domain generalization gap:** The differences between in-domain and cross-domain accuracy, showing how much performance degrades due to domain shift

These metrics allow us to assess both the robustness of our domain generalization techniques as well as which models overfit to source domains as data becomes scarcer.

## References


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